

The Clebsch orientation cubic: arithmetic covers and icosahedral harmonics

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Abstract

The Clebsch matching geometry carries a two-valued orientation whose first odd observable is cubic. We relate this finite orientation to Hitchin’s ordered-icosahedron double cover. The paper’s central target identifies the golden Clebsch torsor with the fibre of the quadratic cover $w^2 = 5J_0$, and identifies its reduction modulo 11 with the Clebsch sheet exchange. An exact tensor calculation already shows that the signed matching cubic and the Clebsch cubic span the same invariant line over $\mathbf{F}_{11}$. The remaining arithmetic specialization and integral comparison are isolated as explicit certification gates. In characteristic zero, the same cubic is the restriction of the standard $\ell = 6$ Gaunt cubic to the four-dimensional icosahedral face-axis channel. A further target asks whether the same real quadratic field is recovered from the relative position of the two icosahedral multiplicity structures in the intermediate Jacobian of the Klein cubic.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; arithmetic double cover; Klein cubic; intermediate Jacobian.

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1 Introduction

The Clebsch matching configuration has two sheets. Forgetting the sheet retains the unordered matching data; recovering it requires an odd observable. Paper II constructs such an observable over $\mathbf{F}_{11}$: a signed cubic tensor on a ten-dimensional matching quotient. The question here is arithmetic. Does this finite orientation arise as a specialization of a characteristic-zero double cover?

Hitchin’s ordered-icosahedron incidence supplies the geometric cover. Its rational function field has the form

$$w^2 = 5J_0.$$

On the Clebsch four-space the invariant J_0 restricts to $16\sigma_3^2$. Away from $\sigma_3 = 0$, division by $4\sigma_3$ then leaves the constant quadratic torsor

$$\mathrm{Spec} \mathbf{Q}(\sqrt{5}) \longrightarrow \mathrm{Spec} \mathbf{Q}.$$

The factor 5 records the orientation field; the cubic records the first place at which orientation becomes visible.

Principal target. The central theorem will identify the golden Clebsch orientation torsor with this Hitchin fibre and prove that its deck exchange reduces modulo 11 to the finite sheet involution T_{11} . This statement is partitioned in the trust ledger as HC-1, HC-2, HC-3, and HC-4. The tensor clause HC-3 is certified, and the rational cover and specialization clauses are proved. The geometric incidence comparison in positive characteristic is asserted only outside an unspecified finite set of primes; the present integral equations do not justify the sharper claim $p \neq 2, 5$.

Harmonics. The axes through opposite icosahedral faces give a canonical ten-dimensional subspace of the degree-six spherical harmonics. Its four-dimensional summand is the Clebsch module, and the standard Gaunt cubic restricts there to a nonzero multiple of σ_3 . This identifies the orientation observable with a symmetry-resolved Steinhardt-type cubic, not merely with an analogous invariant.

Proof architecture. Section 2 separates the general quadratic-cover mechanism from its invariant-theoretic input. Section 3 proves the finite cubic bridge. Section 4 isolates the exact integral specialization still required. Section 5 constructs the harmonic channel. The final sections admit only consequences of these theorems and state the proof modes used.

2 Orientation as a quadratic cover

Let R be a domain with an involution ι . Write

$$R^+ = \{r : \iota(r) = r\}, \quad R^- = \{r : \iota(r) = -r\}.$$

The elementary algebra behind the construction is that one nonzero odd element trivializes the odd summand after localization.

Lemma 2.1 (Localized odd generator). **ALG-1.** *Assume that 2 is invertible in R , and let $C \in R^-$ be nonzero. After localizing at C^2 ,*

$$R[(C^2)^{-1}] = R^+[(C^2)^{-1}] \oplus C R^+[(C^2)^{-1}].$$

Proof. Every $r \in R$ has the decomposition

$$r = \frac{r + \iota(r)}{2} + \frac{r - \iota(r)}{2}.$$

The first term is invariant. If r is odd, then r/C is invariant after C is inverted, because both numerator and denominator change sign. Since inverting C^2 also makes C invertible, the odd summand is C times the invariant summand. The intersection is zero because an element in it is both invariant and odd. $\square$

Thus an orientation-forgetting quotient is generically quadratic whenever its odd algebra is generated by one element. For the sign-twisted icosahedral action on the Clebsch four-space, the first candidate odd generator is the cubic

$$\sigma_3(y) = \frac{1}{3} \sum_{i=1}^5 y_i^3, \quad \sum_{i=1}^5 y_i = 0.$$

Here is the precise characteristic boundary. Put

$$A_R = R[y_1, \dots, y_5]/(e_1), \quad \Delta = \prod_{i < j} (y_i - y_j),$$

where $R = \mathbf{Z}[1/30]$. The alternating group permutes the coordinates, and an odd permutation τ defines the twisted involution $\kappa(y_i) = -y_{\tau(i)}$. Nonmodular invariant theory gives

$$A_R^{A_5} = R[e_2, e_3, e_4, e_5, \Delta]/(\Delta^2 - \text{disc}(e_2, e_3, e_4, e_5)).$$

Indeed, an A_5 -invariant polynomial is the sum of a symmetric polynomial and Δ times a symmetric polynomial; exactness of invariants after imposing $e_1 = 0$ uses $60 \in R^\times$. Moreover

$$\kappa(e_i) = (-1)^i e_i, \quad \kappa(\Delta) = -\Delta.$$

Thus $e_3 = \sigma_3$ is the first nonconstant odd invariant. On $D(e_3)$, Lemma 2.1 identifies the full invariant algebra with its even part plus e_3 times its even part. This statement does not assert the same presentation in characteristics 2, 3, 5; modular invariants can appear there. Hitchin's binary-sextic realization has the same nonmodular boundary: the classical generators and the sextic J_6 used below are available uniformly in characteristic greater than 5 [2, 4]. Dye's earlier finite-geometry theorem already gives the complementary field-level criterion: Clebsch hexagons exist precisely when the characteristic is not 2 and 5 is a square [3, Theorem 1].

Theorem 2.2 (Rational Hitchin–Clebsch cover). *HC-1. Let H be the rational seven-space of harmonic cubics, and let $\mathcal{I}_{\mathbf{Q}}$ be the incidence variety of pairs $([f], U)$, where U is a Mukai–Umemura isotropic three-plane and $f \perp U$. The induced extension of $\mathbf{Q}(\mathbf{P}(H))$ is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}).$$

On the Clebsch four-space,

$$J_0 = 16\sigma_3^2,$$

so over $D(\sigma_3)$ the cover is the constant golden torsor $\text{Spec } \mathbf{Q}(\sqrt{5})$.

Proof. Hitchin realizes the Mukai–Umemura threefold as the zero locus of the three skew forms on $\text{Gr}(3, H)$. The top Chern number of the dual universal bundle is 2, so a general f is orthogonal to two isotropic three-planes [2, §§4–5]. The incidence variety is integral, hence its generic function field is a quadratic extension. Hitchin identifies its branch hypersurface with the irreducible sextic $J_0 = 0$, and computes $J_0|_{\mathbf{P}(V)} = 16\sigma_3^2$ [1, §§9–10]. Since $\text{Pic}(\mathbf{P}(H)) = \mathbf{Z}$ has no two-torsion, the extension therefore has the form $\mathbf{Q}(\mathbf{P}(H))(\sqrt{cJ_0})$ for one rational square class c .

For $f = xyz$, Hitchin exhibits the two icosahedra with golden coordinates $t^2 - t - 1 = 0$. The value $J_0(f)$ is a nonzero rational square by the displayed Clebsch restriction. Hence this fibre determines $c = 5$. Dividing by $4\sigma_3$ gives the final assertion. $\square$

Integral boundary. Choose a full lattice $\Lambda \subset H$ and an integer $a \neq 0$ such that $J_{\mathbf{Z}} = a^2 J_0$ has integral coefficients. The algebra

$$\mathcal{O}_{\mathbf{P}(\Lambda)} \oplus \mathcal{O}_{\mathbf{P}(\Lambda)}(-3), \quad z^2 = 5J_{\mathbf{Z}},$$

is finite locally free of degree two over $\mathbf{Z}$ and has the function field of Theorem 2.2. On $D(10J_{\mathbf{Z}})$ it is finite étale. The primes 2 and 5 are forced bad primes for an ordinary two-sheet interpretation: in characteristic 2 the equation is inseparable, while in characteristic 5 it is generically $z^2 = 0$.

This abstract integral algebra is not yet an integral incidence theorem. The constructions in the cited Mukai–Umemura and Hitchin sources are over $\mathbf{C}$, and the available binary-sextic formulae justify their classical invariant presentation only after inverting 30.

Spreading out the rational incidence isomorphism gives some $\mathbf{Z}[1/N]$, with $2, 3, 5 \mid N$ after enlarging N , but neither the sources nor the present equations determine the other prime divisors or prove that there are none. Consequently every finite-field icosahedral-incidence interpretation in this paper remains restricted to an unspecified cofinite set of primes. The formula for the abstract quadratic algebra itself has no such extra exception.

3 The Clebsch cubic bridge

Let G be the order-60 stabilizer of the H_3 -invariant matching in $\mathrm{PGL}_2(\mathbf{F}_{11})$, and let Ω be its 22-point matching orbit. For a reference matching M_0 , put

$$W = \langle q_M - q_{M_0} : M \in \Omega \rangle.$$

The two $\mathrm{PSL}_2(\mathbf{F}_{11})$ -sheets define signs $\epsilon(M) \in \{\pm 1\}$ and the symmetric tensor

$$T = \sum_{M \in \Omega} \epsilon(M) [q_M - q_{M_0}]^{\otimes 3} \in \mathrm{Sym}^3(W).$$

Theorem 3.1 (Finite tensor bridge). *There is a G -equivariant isomorphism*

$$S : \mathbf{F}_{11}^{\binom{[5]}{2}} \longrightarrow W$$

such that, after using the standard pairing to identify the permutation module with its dual and restricting along

$$E(y)_{\{i,j\}} = y_i + y_j, \quad \sum_i y_i = 0,$$

one has

$$E^*(S^{-1})^{\otimes 3} T = 4\sigma_3.$$

In particular, the signed matching cubic and the Clebsch cubic span the same nonzero invariant line over $\mathbf{F}_{11}$.

Proof. Conjugation on the normalizers of the five Klein four subgroups identifies G with A_5 in its degree-five action. Hence it acts on the ten two-subsets. The matching construction gives a second ten-dimensional representation on W . Solving $\rho_W(g)S = S\rho_P(g)$ produces an invertible intertwiner, and the recorded matrix satisfies this identity for all 60 elements.

Transporting the 220 symmetric coordinates of T through this intertwiner and contracting with E gives a polarized 4^3 -tensor whose entries are 0 when all indices agree and 6 otherwise. The polarization of σ_3 has entries 0 and $-1/3 = 7$, respectively. Since $4 \cdot 7 = 6$ in $\mathbf{F}_{11}$, the two tensors agree. A primary exact calculation and an independent reconstruction verify the representation, intertwiner, and all 64 contracted entries. The final contraction is also checked in Lean. $\square$

Remark 3.2 (Normalization). The characteristic-zero Gaunt scalar presently available has denominator divisible by 11. The theorem therefore identifies a common integral cubic line; it is not obtained by reducing that rational scalar modulo 11.

4 Arithmetic specialization

The arithmetic exchange is already visible on the cubic xyz . Its two ordered icosahedra are Galois conjugate over $\mathbf{Q}(\sqrt{5})$; modulo 11 they become two rational Clebsch parents, and the Galois exchange becomes the nontrivial spinor class. We prove these statements before introducing the Mathieu carrier.

4.1 The golden fibre

Let $t^2 - t - 1 = 0$, and put

$$I_t = \{[0 : t : 1], [0 : t : -1], [1 : 0 : t], [-1 : 0 : t], \\ [t : -1 : 0], [-t : -1 : 0]\} \subset \mathbf{P}^2.$$

These six points represent the six opposite vertex pairs of an icosahedron.

Proposition 4.1 (Golden fibre). *Over a field of characteristic different from 2 and 5, the points of I_t lie on $xyz = 0$, have a common nonzero norm, have normalized squared inner product $1/5$, and form a six-arc. The two configurations over $\mathbf{Q}$ are therefore defined over*

$$\mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$$

and are exchanged by its nontrivial automorphism. In Hitchin's normalized cover of ordered icosahedra they are the complete fibre over $[xyz]$.

Proof. Every displayed vector has one zero coordinate. Since $t^2 = t + 1$, its squared norm is $t^2 + 1 = t + 2$. The inner product of two distinct representatives is $\pm t$, while

$$(t + 2)^2 = 5(t + 1) = 5t^2.$$

This gives the normalized squared inner product $1/5$. The twenty three-point determinants are $\pm 2t$ or $\pm 2t^2$, so none vanishes under the stated hypotheses. The final assertion uses Hitchin's classification of the two icosahedral sets on the orthogonal-plane cubic. $\square$

4.2 The exchanger and the tetrahedral hinge

Set

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The relation $1 - t = -1/t$ gives, by substitution,

$$R(I_t) = I_{1-t}, \quad R^2 = \text{diag}(1, -1, -1), \quad R^4 = 1.$$

Moreover $R(xyz) = -xyz$: it fixes the projective cubic and exchanges its two ordered icosahedra.

Proposition 4.2 (Tetrahedral hinge). *Let $G_t = \text{Stab}_{\text{SO}_3}(I_t) \cong A_5$. Then*

$$RG_tR^{-1} = G_{1-t}, \quad G_t \cap G_{1-t} \cong A_4.$$

Proof. The first equality follows from $R(I_t) = I_{1-t}$. A direct linear calculation shows that the cubics through $I_t \cup I_{1-t}$ form the one-dimensional space spanned by xyz : write a general ternary cubic as

$$ax^3 + bx^2y + cx^2z + dxy^2 + exyz + fxz^2 + gy^3 + hy^2z + iz^2 + jz^3.$$

Substitution of the three opposite pairs, followed by $t^2 = t + 1$, gives

$$\begin{aligned} g = i = 0, \quad h(t + 1) + j = 0, \quad c + j(t + 1) = 0, \\ a + f(t + 1) = 0, \quad b(t + 1) + g = 0, \quad a(t + 1) + d = 0. \end{aligned}$$

Equating the rational and t -parts kills every coefficient except e . Thus every element of the intersection fixes the plane-triple cubic.

The rotations given by even signed coordinate permutations fix xyz and each I_t ; they form the tetrahedral group A_4 . Conversely, the icosahedral group $G_t \cong A_5$ acts on its five plane-triple cubics. This is its standard transitive action of degree five, whose point stabilizer is A_4 . Hence the common stabilizer is exactly the displayed tetrahedral subgroup. $\square$

4.3 Reduction modulo eleven

The roots of $t^2 - t - 1$ in $\mathbf{F}_{11}$ are 4 and 8. For the working Clebsch arc

$$A = \{[1 : 10 : 0], [1 : 9 : 1], [1 : 4 : 7], \\ [1 : 8 : 5], [0 : 1 : 4], [1 : 1 : 7]\},$$

define

$$M_4 = \begin{pmatrix} 1 & 4 & 7 \\ 5 & 10 & 3 \\ 1 & 8 & 1 \end{pmatrix}, \quad M_8 = \begin{pmatrix} 4 & 5 & 5 \\ 9 & 10 & 7 \\ 4 & 7 & 10 \end{pmatrix}.$$

Six projective substitutions give

$$M_4 I_4 = A, \quad M_8 I_8 = A,$$

and $\det M_4 = 1$, $\det M_8 = 9$. Furthermore,

$$M_8^{-1} M_4 = 3R$$

in $\mathrm{GL}_3(\mathbf{F}_{11})$. Thus forgetting the ordered parent identifies both configurations with the same unmarked Clebsch arc, while their relative projectivity is the reduction of the golden exchanger.

Proposition 4.3 (Spinor specialization). *The reduction of R represents the nontrivial element of*

$$T_{11} = \mathrm{PGL}_2(11)/\mathrm{PSL}_2(11).$$

Proof. For $Q(x, y, z) = x^2 + y^2 + z^2$, reflection in a vector v has spinor class $Q(v)$. On the yz -plane,

$$R = s_{e_2} s_{e_2 - e_3},$$

so

$$\theta(R) = Q(e_2)Q(e_2 - e_3) = 2 \quad \text{in } \mathbf{F}_{11}^\times/\mathbf{F}_{11}^{\times 2}.$$

The class of 2 is nonsquare modulo 11. The standard identifications

$$\mathrm{SO}_3(11) \cong \mathrm{PGL}_2(11), \quad \Omega_3(11) \cong \mathrm{PSL}_2(11)$$

identify the spinor quotient with T_{11} , proving the claim. $\square$

4.4 The marked Mathieu carrier

The ternary Golay–Hadamard model supplies two explicit subgroups

$$H_{\text{row}}, H_{\text{col}} \cong M_{11} \quad \text{inside} \quad M_{12}.$$

One is the pure coordinate group of the signed code and the other is the stabilizer of the parity coordinate. Their exact twelve-point generators give

$$H_{\text{row}} \cap H_{\text{col}} = \text{PSL}_2(11).$$

Consequently

$$|H_{\text{row}}H_{\text{col}}| = \frac{7920^2}{660} = 95040 = |M_{12}|,$$

so the two parents generate M_{12} . Hadamard row–column duality induces the noninner automorphism of M_{12} and exchanges their two conjugacy classes. The finite assertions in this paragraph are checked from the displayed permutation carriers by the primary certificate and an implementation-independent replay described in Section 7.

There is no canonical unmarked identification between the golden and Mathieu pairs. The precise conclusion is the following marked statement.

Corollary 4.4 (Marked Hitchin–Mathieu torsor). *Choose the marking $I_4 \leftrightarrow H_{\text{row}}$. There is a unique equivariant bijection*

$$\{I_4, I_8\} \longrightarrow \{H_{\text{row}}, H_{\text{col}}\}$$

that extends this choice. It sends I_8 to H_{col} , and intertwines the reduced Hitchin exchanger with Hadamard row–column duality. Without the marking there are exactly two such equivariant bijections and neither parent is distinguished.

Proof. Both sets are free transitive C_2 -sets: Proposition 4.3 supplies the nontrivial exchange on the Hitchin side, and Hadamard duality supplies it on the Mathieu side. The image of one marked element determines an equivariant map between free transitive C_2 -sets. There are two possible markings. $\square$

This marked, and only marked, compatibility is HC-5.

The certificate for this section checks only the explicit substitutions, matrix identities, permutation-group orders, intersection, and carrier exchange. Propositions 4.1–4.3 and Corollary 4.4 are the mathematical proof; the certificate is not used to infer the golden field, the A_4 hinge, the spinor formula, or the marked-torsor lemma.

Integral boundary. The abstract quadratic cover gives, on its nonbranch locus,

$$\#X_f(\mathbf{F}_q) = 1 + \chi_q(5J(f)).$$

Its geometric interpretation over finite fields currently holds only away from an unspecified finite set of primes. The final theorem will name the largest justified integral base supplied by an explicit model; it will not replace that finite exceptional set by $p \neq 2, 5$ without proof.

5 The degree-six harmonic realization

Let $u_e \in S^2$, $1 \leq e \leq 10$, be the unoriented axes through pairs of opposite icosahedral faces. The zonal degree-six harmonics define

$$L : \mathbb{R}^{10} \longrightarrow \mathcal{H}_6, \quad L(a)(\omega) = \sum_e a_e P_6(u_e \cdot \omega),$$

where

$$P_6(s) = \frac{1}{16} (231s^6 - 315s^4 + 105s^2 - 5).$$

The degree-six bond-orientational invariant is the first standard icosahedral channel in the Steinhardt–Nelson–Ronchetti construction [7, 8]. The following theorem identifies its four-dimensional symmetry-breaking component.

Theorem 5.1 (Harmonic Clebsch bridge). *PH-1, PH-2, and PH-3. Let A be the Petersen adjacency matrix on the ten face axes.*

1. *The map L is injective, and its Gram matrix is*

$$K = \frac{1}{243}(196I + 47J - 112A).$$

Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1} \oplus V_4 \oplus V_5$ are respectively

$$\frac{110}{81}, \quad \frac{140}{81}, \quad \frac{28}{81}.$$

2. *Label the axes by the pairs $\{i, j\} \subset \{1, \dots, 5\}$. The Clebsch summand is the Petersen (-2) -eigenspace*

$$a_{ij} = y_i + y_j, \quad \sum_i y_i = 0.$$

3. *For*

$$F_y(\omega) = \sum_{i < j} (y_i + y_j) P_6(u_{ij} \cdot \omega),$$

the spherical cubic is

$$\frac{1}{4\pi} \int_{S^2} F_y(\omega)^3 d\omega = -\frac{784000}{1247103} \sigma_3(y).$$

If $F = \sum_{m=-6}^6 q_{6m} Y_{6m}$ in the orthonormal Condon–Shortley convention and

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1} q_{6m_2} q_{6m_3},$$

then

$$\int_{S^2} F^3 d\omega = -\frac{130}{\sqrt{3553}\pi} W_6(F).$$

Proof. The face-axis permutation representation decomposes as $\mathbf{1} \oplus V_4 \oplus V_5$, while

$$\mathcal{H}_6|_{A_5} \simeq \mathbf{1} \oplus V_3 \oplus V_4 \oplus V_5.$$

Direct evaluation of $P_6(u_e \cdot u_f)$ gives the displayed Gram matrix. The Petersen eigenvalues 3, 1, -2 , with multiplicities 1, 5, 4, give the three positive Gram eigenvalues and prove injectivity. For $a_{ij} = y_i + y_j$,

$$(Aa)_{ij} = \sum_{\{k,l\} \cap \{i,j\} = \emptyset} (y_k + y_l) = -2(y_i + y_j),$$

so these vectors form the four-dimensional eigenspace.

The cubic integral is A_5 -invariant on V_4 . The invariant cubic line is generated by σ_3 , so one nonzero value determines the scalar. At $y = (4, -1, -1, -1, -1)$, exact spherical moments give

$$\frac{1}{4\pi} \int F_y^2 = \frac{2800}{351}, \quad \frac{1}{4\pi} \int F_y^3 = -\frac{15680000}{1247103},$$

while $\sigma_3(y) = 20$. This proves the third displayed scalar.

Finally, the Gaunt formula expresses the triple integral in the orthonormal spherical-harmonic basis as a product of two Wigner $3j$ -symbols. Here

$$\begin{pmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{pmatrix} = -\frac{20}{\sqrt{46189}},$$

which gives the stated conversion to W_6 . The exact axis, Gram, moment, and Wigner calculations are checked by two independent implementations. $\square$

Order-parameter meaning. A uniform icosahedral neighbour cage lies in the trivial summand. The Clebsch cubic instead sees the V_4 component of nonuniform weights on the ten opposite face pairs. For a face-axis weight vector a , the projector

$$P_{V_4} = \frac{(A - 3I)(A - I)}{15}$$

extracts this component. A scale-free signed descriptor is then

$$C_4(a) = \frac{\langle L(P_{V_4}a)^3 \rangle}{\langle L(P_{V_4}a)^2 \rangle^{3/2}}, \quad \langle f \rangle = \frac{1}{4\pi} \int_{S^2} f.$$

For the witness in the proof,

$$C_4 = -\frac{120\sqrt{273}}{3553}.$$

Possible weights include opposed-face distortion, chemical decoration, or normal stress. Whether C_4 predicts rearrangement, yielding, or nucleation beyond standard descriptors is an empirical question, recorded as PH-4. No applied performance claim is made here.

Remark 5.2 (Which symmetry flips the sign). The cubic is invariant under the rotational A_5 -action. The sheet exchange is the odd element in the sign-twisted extension of this action, not an ordinary rotation of a uniformly weighted icosahedron. Any physical realization must therefore specify the decorated face-axis state and the outer-odd operation; the existence of an icosahedral cage alone does not produce the bit.

6 Consequences and boundaries

Once the arithmetic specialization is proved, three consequences become available without creating independent narratives.

1. The cubic-first orientation in the finite matching quotient becomes the reduction of a characteristic-zero quadratic orientation cover.
2. The two golden icosahedral parents become the two fibres of one arithmetic torsor, with the A_4 intersection as their common hinge.
3. The marked Hitchin–Mathieu correspondence becomes a corollary after the two carrier identifications are verified.

Theta, Fourier, quantum, four-sheet holonomy, and degree-23 coherence comparisons are not part of the present theorem complex. They enter only if they can be derived from the orientation cover with precise hypotheses. Otherwise they remain research inventory.

7 Verification and trust

The proof surface distinguishes conceptual arguments, classical inputs, exact certificates, independent replays, and Lean terminals. The machine-readable ledger in `verification/trust_manifest.json` records the current status of every manuscript claim.

The finite tensor bridge uses exact arithmetic in two implementations. The primary route exports the 22 orbit vectors, the sheet signs, the ten-dimensional intertwiner, and the contracted tensor. The independent route reconstructs the group, orbit, quotient, and contraction. Lean checks the final literal 4^3 -tensor equality over $\mathbf{F}_{11}$; it does not formalize the upstream orbit construction or all 60 equivariance equations.

The harmonic bridge likewise has a primary exact generator, a canonical JSON certificate, and a separate replay. Both reconstruct the ten face axes over $\mathbf{Q}(\sqrt{5})$, the full Gram and Petersen matrices, and the quadratic and cubic spherical moments. The replay separately checks the Petersen minimal polynomial and the factorial formula for the Wigner $3j$ -symbol. Neither route uses floating-point arithmetic.

The arithmetic specialization is proved in the text. Its compact certificate checks only the six projective substitutions, the comparison and reflection matrices, and the explicit twelve-point Mathieu carriers. The independent replay reimplements those checks and enumerates the two M_{11} parents, their order-660 intersection, their M_{12} join, and the carrier exchange without importing the primary program. Neither route is used to infer the golden field, the A_4 hinge, the spinor formula, or the marked-torsor lemma. The exact commands are

```
python3 verification/evidence/arithmetic_cover.py --check
python3 verification/evidence/arithmetic_cover_replay.py
sha256sum -c verification/evidence/arithmetic_cover.sha256
```

run from the paper directory. The integral model and novelty claims are established by the source audit and the scheme-theoretic argument in Section 2, not by a certificate. That argument proves the rational 5-twist and the abstract integral quadratic algebra, while retaining an unspecified finite bad set for comparison with the geometric incidence model.

The current verification runner checks manuscript structure and trust-ledger coverage only. It intentionally refuses to report release readiness while any load-bearing claim remains gated.

8 Conclusion

The finite Clebsch orientation is already known to be cubic-first: its signed matching tensor restricts to the Clebsch cubic line over $\mathbf{F}_{11}$. The principal remaining problem is to show

that this line is the specialization of Hitchin’s arithmetic double cover and that its deck exchange is exactly the finite sheet involution.

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